Zachary van Ryn

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SUMMARY

Computer Science Games student at USC with experience building C++/SDL2 games on a shared Actor/Component engine, including A* enemy AI and 3D rendering; additionally created Python automation tools used by university staff.

EDUCATION

University of Southern California, Viterbi School of Engineering Los Angeles, CA

Bachelor of Science in Computer Science (Games) Expected May 2027

Relevant Coursework: Video Game Programming, Linear Algebra and Linear Differential Equations, Algorithms and Theory of Computing

RELEVANT PROJECT EXPERIENCE

Video Game Programming, USC

C++ / SDL2 Game Development August 2026 – Present

- Built 7 games in C++/SDL2, from Pong to a Star Fox-style 3D shooter, on one shared Actor/Component engine
- Implemented engine systems for collision detection, sprite animation, tilemaps, audio, and 3D rendering
- Programmed A* pathfinding for ghost AI in a Pac-Man clone; built Frogger, Mario, and Zelda-style games

***Dreamcatchers* Game Project** August 2026 – Present

Usability Team

- Co-developing a game with a team; Plan and run observed think-aloud playtests to guide design
- Wrote up playtest findings and reported them back to all teams to shape each design iteration

***Eagles Nest* Unity Game Project C#** January 2026 – May 2026

- Developed a 2D top-down game in Unity with a complete narrative, enemy pathfinding AI, and puzzle mechanics
- Programmed a key-and-door puzzle system in C# that unlocks doors when a key is brought to a target tile
- Designed progressively complex puzzles and story elements from playtest feedback
- Took the game through full preproduction, from pitch and prototype to vertical slice and game trailer

Game Design Workshop

Board Game Producer August 2022 – December 2022

- Accounted for a $35 budget, cutting expensive mechanics and components to stay within scope
- Recorded progress via a playtesting report and photo design journal

WORK EXPERIENCE

USC Student Account Services Los Angeles, CA

Student Worker June 2024 – Present

- Extended an unfinished Python toolkit that sorts and routes bank statement PDFs into SharePoint for 41 accounts
- Prevented misfiled records by detecting mislabeled file formats and cross-checking account numbers against PDF text
- Packaged the toolkit with one-click launchers and a setup script so non-technical coworkers can run it without help
- Process student checks and payments through the university's financial system

Payroll Assistant, USC Rossier School of Education August 2023 – May 2024

Delivery Driver, Domino's April 2022 – August 2023

Brand Representative, Abercrombie and Fitch April 2021 – April 2022

SKILLS

Programming Languages: C++, C#, Python, JavaScript

Game Development: Unity, SDL2, Actor/Component architecture, A* Pathfinding, Collision Detection, 3D Rendering, Playtesting, Usability, Q/A

Software & Tools: GitHub, SOLIDWORKS, SharePoint/OneDrive, OBS Studio